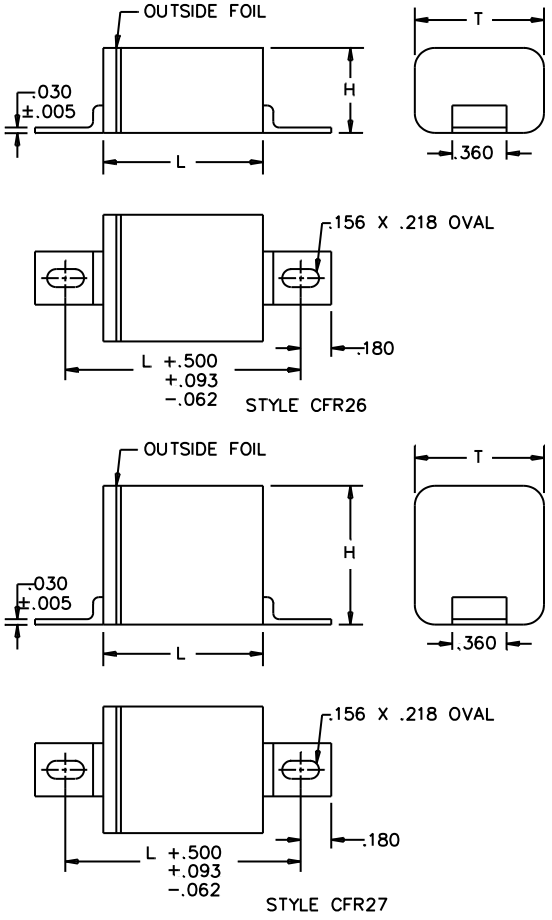
INCH-POUND
MIL-PRF-55514/11C
w/AMENDMENT 2
17 September 2018
SUPERSEDING
MIL-PRF-55514/11C
w/AMENDMENT 1
15 August 2002

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, FIXED, METALLIZED PLASTIC DIELECTRIC, DC, IN NONMETAL CASES, ESTABLISHED RELIABILITY, STYLES CFR26 AND CFR27

The specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55514.



Inches	mm
.002	0.05
.005	0.13
.030	0.76
.062	1.57
.093	2.36
.156	3.96
.180	4.57
.218	5.54
.360	9.14
.500	12.70

NOTES:
1. Dimensions are in inches. Metric equivalents are given for general information.
2. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
3. These capacitors are to be mounted by screws.

FIGURE 1. Styles CFR26 and CFR27 capacitors.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Non-hermetic.

Dielectric: Metallized polycarbonate (characteristic R); or Polyphenylene sulfide (characteristic U).

Characteristic: R or U as specified in MIL-PRF-55514.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Solderability: Not applicable.

Operating temperature range: -55°C to +85°C, when properly voltage derated (see figure 2), characteristic R or U capacitors will operate to +125°C.

Voltage rating: See table I.

Preconditioning: Not applicable.

Burn-in: In accordance with MIL-PRF-55514, except for the following conditions:

DC burn-in voltage shall be 140 percent of the dc rated voltage (see table I) applied for 48 hours minimum at 125°C, +4°C, -0°C.

AC burn-in: Not applicable.

Insulation resistance: Insulation resistance shall be measured at rated voltage (see table I) at the applicable temperature. Charge time 2 minutes, maximum; however, for capacitance values greater than 1.0 μF, an additional 1 minute per μF is permitted. The following limits shall apply:

Temperature	Insulation resistance
+25°C	50,000
+85°C	5,000
+125°C	500

Marking: In accordance with MIL-PRF-55514.

Qualification submittal:

Part or Identifying Number (PIN)	Sample size
CFR27LUA106JM	61
CFR27LUA506JM	61
CFR27LUG506JM	61
CFR27LUB506JM	61

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TABLE I. Capacitor characteristics and dimensions.

PIN 1/	Capacitance (μF)	Capacitance tolerance	Rated voltage at 85°C	ESR milliohms at 100 KHz	Ripple current max at 25°C	Resonant frequency kHz	Style CFR26			Style CFR27		
							T 2/ 4/	H 2/ 4/	L 3/ 4/	T 2/ 4/	H 2/ 4/	L 3/ 4/
CFR2-L-A106--	10.0	J, K, M	50 V	11	12.6	265	.812	.562	.750	.562	.812	.750
CFR2-L-A206--	20.0	J, K, M	50 V	10	14.8	178	.968	.718	.875	.718	.968	.875
CFR2-L-A506--	50.0	J, K, M	50 V	6	23.5	108	1.375	1.000	1.000	1.000	1.375	1.000
CFR2-L-G505--	5.0	J, K, M	75 V	13	11.6	356	---	---	---	.500	.750	.875
CFR2-L-G106--	10.0	J, K, M	75 V	12	13.5	251	1.031	.718	.875	.718	1.031	.875
CFR2-L-G206--	20.0	J, K, M	75 V	11	16.5	171	1.312	.937	1.000	.937	1.312	1.000
CFR2-L-G506--	50.0	J, K, M	75 V	6	24.2	100	1.687	1.312	1.250	1.312	1.687	1.250
CFR2-L-B505--	5.0	J, K, M	100 V	13	12.4	356	.906	.593	.875	.593	.906	.875
CFR2-L-B106--	10.0	J, K, M	100 V	12	14.6	242	1.093	.781	1.000	.781	1.093	1.000
CFR2-L-B206--	20.0	J, K, M	100 V	11	17.1	160	1.281	.906	1.250	.906	1.281	1.250
CFR2-L-B506--	50.0	J, K, M	100 V	6	23.6	96	1.812	1.437	1.375	1.437	1.812	1.375

1/ The complete PIN will include additional symbols to indicate style (6 for CFR26 or 7 for CFR27), characteristic (U or R), capacitance tolerance, and product level (C, M, P, R and S).
2/ Tolerances: Thickness and length +.062 inch, -.156 inch (+1.57 mm, -3.96 mm) on .500 inch (12.70 mm) and +.093, -.187 (+2.36 mm, -4.75 mm) on .562 inch (14.27 mm) and greater.
3/ Tolerances: Length +.093 inch, -.062 inch (+2.36 mm, -1.57 mm).
4/ See dimension table below:

Inches	mm	Inches	mm	Inches	mm
.500	12.70	.875	22.23	1.250	31.75
.562	14.27	.906	23.01	1.281	32.54
.593	15.06	.937	23.80	1.312	33.32
.718	18.24	.968	24.59	1.375	34.93
.750	19.05	1.000	25.40	1.437	36.50
.781	19.84	1.031	26.19	1.687	42.85
.812	20.62	1.093	27.76	1.812	46.02

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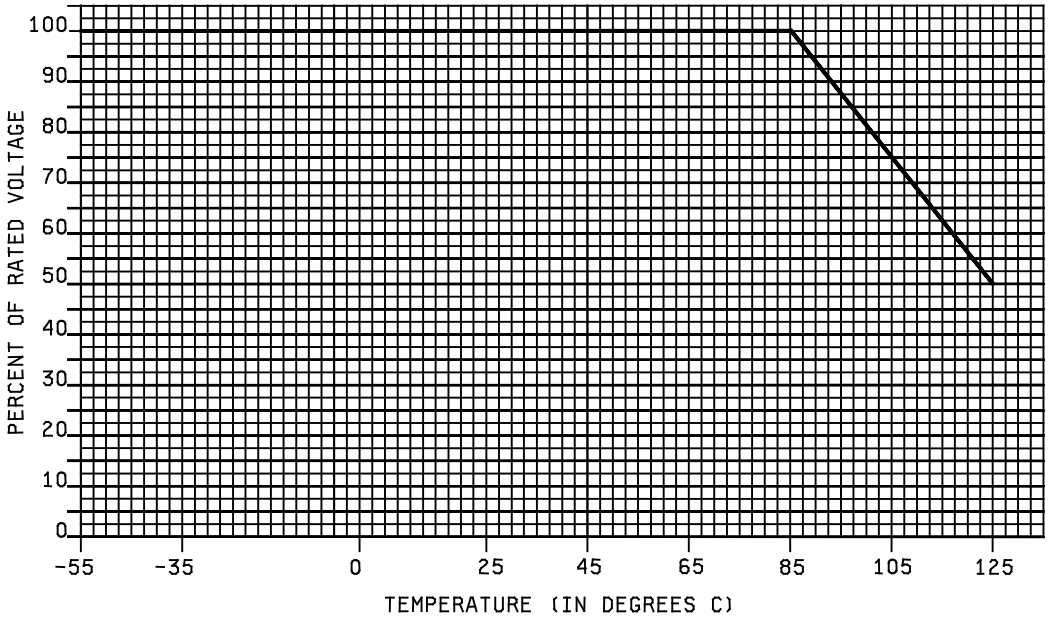


FIGURE 2. Voltage derating with temperature.

Amendment notations: The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
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Review activities:
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

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(Project 5910-2018-036)

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